DhwaniLab — Telugu ASR Benchmarking

Phase 1 report · three pretrained AI4Bharat models on IndicVoices Telugu · WER and OI-WER

1. Objective

Benchmark publicly released Telugu ASR models under a single identical evaluation protocol, and evaluate them with both standard Word Error Rate (WER) and Orthographically-Informed WER (OI-WER), so that reported differences reflect the models rather than the measurement. This is Phase 1; Phase 2 will build a new Telugu dataset and fine-tune these models on it.

2. Experimental setup

Component	Choice
Dataset	ai4bharat/IndicVoices, config telugu, split valid (streamed)
Size	3,295 utterances / 37,350 reference words
Reference field	normalized
Audio	mono, resampled to 16 kHz
Language model	none, for either model
Output	per-utterance CSV: index, speaker_id, duration, reference, prediction, S/D/I

The validation split is the only evaluation split IndicVoices publishes; there is no separate test split (train = 269,300 utterances, valid = 3,295).

Models evaluated

- **IndicWhisper** (`whisper-medium-te_alldata_multigpu`) — standard HuggingFace checkpoint, decoded autoregressively with language=te, task=transcribe.
- **IndicConformer** (`ai4bharat/indic-conformer-600m-multilingual`) — gated repository, loaded with trust_remote_code; resolves to an ONNX Runtime model, decoded with greedy CTC.
- **IndicWav2Vec** (`te.pt`, 3.53 GB Fairseq checkpoint) — no HuggingFace Telugu weights exist, so it runs in an isolated Python 3.10 / torch 1.13 virtual environment via a subprocess decoder, with audio pre-dumped to disk. Greedy CTC (Viterbi), no language model.

3. How WER is computed

3.1 Per utterance

Each utterance is scored independently. Both the reference and the prediction are whitespace-normalized and split into word tokens. The two token sequences are then aligned using **minimum edit distance** (word-level Levenshtein alignment), which finds the alignment requiring the fewest edits. Each aligned position is classified as:

- **Hit** — reference word and predicted word are identical.
- **Substitution (S)** — a reference word was replaced by a different predicted word.
- **Deletion (D)** — a reference word has no counterpart in the prediction (word missed).
- **Insertion (I)** — a predicted word has no counterpart in the reference (word invented).

3.2 Corpus aggregation

Per-utterance counts are summed across all 3,295 utterances *before* dividing. The reported figure is therefore a corpus-level rate:

```
WER = ( ΣS + ΣD + ΣI ) / ΣN
where N = number of reference words in an utterance
```

This is deliberately not the mean of per-utterance WERs. Averaging per-utterance rates gives a one-word utterance the same weight as a forty-word utterance, which inflates the influence of short utterances and produces numbers that are not comparable with published results. Per-utterance WER is still stored in the CSVs, but it is used only for error analysis, never for the headline figure.

3.3 Ensuring the two models are comparable

- Both models were run over the same streamed split, so utterance *index* aligns across both result files.

- References were compared position by position between the two runs: **3,295 of 3,295 matched exactly.** Had they differed, the two WERs would not have been measuring the same task.
- Both models are scored against the identical denominator of 37,350 reference words.
- A separate notebook recomputes both WERs from the raw stored predictions; it reproduced the original IndicWhisper figure exactly, confirming the harness itself introduces no difference.

4. How OI-WER is computed

OI-WER addresses a known problem in Indian-language ASR evaluation: many predictions marked wrong are in fact legitimate alternative spellings. The published method (arXiv:2603.00941) replaces each single reference word with a **set of acceptable variants**; a prediction matching any variant is counted as a hit rather than a substitution.

4.1 Algorithm

The reference is represented as an ordered sequence of *segments*, each carrying one or more accepted token sequences. Alignment then runs as a dynamic program over (segment index, prediction position), where each segment may be matched by any of its variants — and a variant may span multiple words, so one reference word can legitimately match two predicted words and vice versa. The error counting and corpus aggregation are otherwise identical to Section 3, and **the denominator remains the original reference word count**, so WER and OI-WER stay directly comparable.

4.2 Equivalence rules implemented

Because no LLM API was available, variants are generated by deterministic rules rather than by an LLM. Four rule groups are applied:

- **Inverse Text Normalization (ITN)** — spans of Telugu number words are parsed to their numeric value and the digit form is added as an accepted variant, so ■■■■■■■ and 8 match. The parser handles units, teens, tens, and the multipliers hundred / thousand / lakh / crore, including compound values such as 399,999.
- **Compound merging and splitting** — a reference token matches a group of predicted tokens (or the reverse) when their concatenations are identical. This is pure string equality, requiring no lexicon, and handles Telugu agglutination.
- **Annotation tag removal** — the transcriber marker <unintelligible> (60 utterances) is removed from references, since no model can produce it and it is guaranteed to score as an error.
- **Surface normalization** — Unicode NFC, punctuation and symbol stripping, removal of replacement characters, and case folding for embedded Latin text.

5. Results

5.1 Headline

Model	WER	OI-WER	Gain	Decoding
IndicConformer	**27.84%**	**26.92%**	**-0.91**	**greedy CTC**
IndicWav2Vec	54.56%	52.48%	-2.07	greedy CTC
IndicWhisper	56.74%	54.34%	-2.40	autoregressive
best-worst spread	28.90	27.41	-1.49	

Ranking is identical under both metrics: IndicConformer < IndicWav2Vec < IndicWhisper.

5.2 Rule ablation

Each row adds one rule group cumulatively, showing what each contributes.

Configuration	IndicConformer	IndicWav2Vec	IndicWhisper
WER (baseline)	27.839%	54.560%	56.736%
+ strip <unintelligible>	27.845%	54.576%	56.765%
+ surface normalization	27.845%	54.576%	56.762%
+ ITN numbers	27.839%	54.386%	56.236%
+ merge / split = OI-WER	**26.925%**	**52.485%**	**54.338%**

Removing <unintelligible> raises WER very slightly: the reference word disappears from the denominator while the model's output for that audio becomes an insertion.

5.3 What the rules corrected

The improvement comes almost entirely from substitutions — 232, 522 and 661 removed for IndicConformer, IndicWav2Vec and IndicWhisper respectively — the expected signature, since accepting a variant can only convert a wrong word into a correct one. Two findings stand out:

- **The digit penalty was entirely one-sided.** ITN removed 124 substitutions from IndicWhisper and zero from IndicConformer, because IndicConformer never emits digits while IndicWhisper does so in 129 utterances. Cases such as reference 'eight' against prediction '8' were perfect recognitions previously scored as complete errors.
- **Merge/split was the larger effect**, as expected for an agglutinative language: a single Telugu word written by the model with an internal space was previously counted as two separate errors.

Representative corrections — reference against prediction, identical words differing only in spacing or number format:

హైదరబాద్కి / హైదరబాద్ కి · మూడువేల / మూడు వేల · రైతుభరోసా / రైతు భరోసా · ఎనిమిది / 8

6. Verification of the scoring code

The OI-WER engine was written from scratch, so it was validated before being trusted:

- With trivial single-variant segments it reduces to standard WER, and reproduces both published figures to six decimal places (27.839357% and 56.736278%), matching the independent jiwer-based computation.
- Nine unit tests cover exact match, substitution, deletion, insertion, multi-word variants, numeral variants, and a negative case confirming a genuine error is still counted as an error.
- The invariant OI-WER ≤ WER was verified to hold for every configuration.
- Malformed variant sets are rejected by a validator; the affected utterance falls back to plain WER scoring, so a bad variant can never silently deflate the score.

7. Interpreting the model gap

- **Domain match.** IndicConformer is AI4Bharat's own model, trained on IndicVoices, so this validation split is in-domain for it; IndicWhisper was fine-tuned on a different Telugu mixture and faces a domain shift. The supportable claim is that IndicConformer is stronger *on this benchmark*, not that it is a better Telugu ASR model in general.
- **Decoding asymmetry.** IndicConformer used greedy CTC with no language model; IndicWhisper generates autoregressively with an implicit internal language model.
- **Different failure modes.** IndicWhisper's much higher deletion and insertion counts indicate dropped content and hallucinated repetition on short utterances, rather than uniform mis-recognition.

8. Limitations

- **This is an approximation of OI-WER, not a reproduction of it.** The published method uses an LLM to propose variants followed by native-speaker review, covering seven variation categories. Our rule-based implementation covers two: ITN and compound merging/splitting. The remaining five — matra and diacritic variants, loanword spellings, phonetic and dialectal variants, ligature variants, and sandhi — require a lexicon or an LLM and are not implemented. This is closer to the paper's WER-SN baseline than to full OI-WER, and our gains (0.91 and 2.40 points) are correspondingly smaller than the paper's 6.3-point average.
- **The Telugu number lexicon requires native-speaker review.** It is validated against every numeric example found in the dataset, but an incorrect entry would silently forgive a real error.
- **A local LLM was evaluated for variant generation and rejected.** gemma3:4b, the largest Gemma that fits in 5 GB of VRAM, produced roughly half hallucinated or omitted variants on Telugu — in one case altering the original word rather than offering an alternative spelling — at about 4 seconds per utterance. It was not used. The paper's results rely on Gemini-2.5-Pro followed by native-speaker review.
- **No human-perceived WER reference.** The paper validates OI-WER against post-edited human transcripts; we have no such reference, so we cannot measure how much closer our metric sits to human judgement.

9. Note on IndicWav2Vec decoding

AI4Bharat never published Telugu IndicWav2Vec weights in HuggingFace format — the hosted repository is empty — so the only Telugu artifact is a Fairseq checkpoint tied to Python 3.10 and torch 1.13. It was evaluated by running that stack in an isolated virtual environment, driven as a subprocess, with audio pre-dumped to disk so the old environment never touches the modern data libraries. The run completed all 3,295 utterances at roughly 8 utterances per second.

One caveat matters for comparison with published figures: AI4Bharat's official IndicWav2Vec pipeline decodes with KenLM and a lexicon. That was deliberately skipped here so that all three models are compared without an external language model. Our 54.56% is therefore an LM-free figure and should not be read against published numbers obtained with LM decoding.

A decoding detail worth recording: in Fairseq's CTC convention the blank symbol is the dictionary's bos index, not pad. Filtering the wrong index leaves every blank frame in the output, which renders as literal markup and produces a near-100% error rate that looks like model failure rather than a decoding bug.

10. Next steps

- Confirm whether AI4Bharat's orthographically-informed IndicVoices benchmark (Telugu: ~2.6K utterances, 92.7K variations, native-speaker reviewed) can be obtained directly — this would replace our rule-based approximation with the reference implementation.
- Native-speaker review of the Telugu number lexicon and the equivalence rules.
- Error analysis from the stored predictions: WER against utterance duration, per-speaker breakdown, substitution patterns, and CER alongside WER. No GPU inference required.
- Optionally re-run IndicWav2Vec with KenLM + lexicon decoding to match AI4Bharat's official pipeline.
- Phase 2: Demucs source separation, VAD segmentation, YouTube subtitles, MFA alignment, a 10-hour Telugu dataset, and fine-tuning all three models — re-evaluating with the same WER and OI-WER harness so the improvement from fine-tuning is measurable per model.